

Loan Approval Prediction Using Machine Learning

Project Title: Loan Approval Prediction Using Machine Learning

Course: AI Data Science / Machine Learning Assignment

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1. Introduction & Objective

The main goal of this project is to build a Machine Learning model that can automatically predict whether a loan application should be approved or denied based on the applicant's profile. This automation helps financial institutions speed up the process and minimize the risk of loan defaults.

2. Data Understanding & Preprocessing

- **Dataset Shape:** [614 rows, 13 columns].
- **Missing Values:** Identified missing values in variables like **Credit_History**, **LoanAmount**, and **Gender**. These were imputed using the **Median** for numerical features and **Mode** for categorical features.
- **Categorical Encoding:** Text features (like Gender, Married, Property_Area) were transformed into numbers using **LabelEncoder**.
- **Feature Scaling:** **StandardScaler** was applied to normalize continuous variables like **ApplicantIncome** and **LoanAmount** so they are on the same scale.

3. Exploratory Data Analysis (EDA) Insights

- **Correlation Heatmap:** The heatmap showed that **Credit_History** has the strongest positive correlation with **Loan_Status**.
- **Loan Status Count:** The dataset is slightly imbalanced, with more approved loans ('Y') than denied ones ('N').
- **Income vs Loan Approval:** Applicants with stable incomes and good credit history have a significantly higher chance of loan approval.

4. Model Performance & Comparison

We trained three different machine learning algorithms and evaluated them using Accuracy, Precision, Recall, and F1-Score.

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Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	79%	75%	98%	85%
Decision Tree	69%	75%	77%	76%
Random Forest	75%	74%	92%	82%

5. Conclusion

Based on the final comparison, the **[Logistic Regression]** performed the best with an accuracy of **[79%]**.

This model is recommended for deployment because it handles non-linear relationships and missing values robustly, providing the most reliable loan risk prediction.